GATE ME 90 Days Challenge

Day 90/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Full Syllabus	**Grand Mock (Day 90)**	**7 days to GATE**

How to use this set: 15 questions (8 MCQ, 7 NAT) on a final mixed full-syllabus warm-up for the Grand Mock: antonyms, interior angles of a polygon, a second-order ODE, Simpson's 1/3 rule, friction on an inclined plane, the middle-third rule, Euler buckling about the weaker axis, rack and pinion, torsional vibration, helical spring stiffness, jet power, conduction from a sphere (Nu = 2), nozzle exit velocity, gravity die casting and the EOQ. Try them in **60 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.
Today's test: open 7 to 11 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 65 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

Select the word that is most nearly OPPOSITE in meaning to PROLIFIC.

(A) Productive
(B) Barren
(C) Abundant
(D) Fertile

Q2 NAT

The sum of the interior angles of a convex polygon is 1800°. The number of sides of the polygon is _____

Q3 MCQ

The solution of $y'' - y = 0$ that satisfies $y(0) = 2$ and $y'(0) = 0$ is

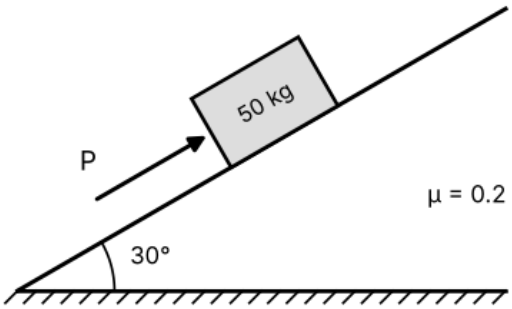
(A) 2 sinh x
(B) 2 cos x
(C) e^x
(D) 2 cosh x

Q4 NAT

The integral of e^{x^2} from x = 0 to x = 1 is evaluated by Simpson's 1/3 rule with two equal intervals (h = 0.5). The result (round off to three decimal places) is _____

Q5 NAT

A 50 kg block is pushed up a rough plane inclined at 30° to the horizontal by a force acting parallel to the plane, as shown. The coefficient of kinetic friction is 0.2. Taking g = 9.81 m/s², the force needed to move the block up the plane at constant speed (in N, round off to one decimal place) is _____



Q6 MCQ

A short column of solid rectangular section, b wide and d deep, carries a compressive load whose line of action is shifted along the depth d. For no tensile stress to develop anywhere in the section, the eccentricity e must not exceed

(A) d/4
(B) d/3
(C) d/6
(D) d/8

Q7 NAT

A steel strut of solid rectangular section 40 mm × 20 mm and length 1 m is pinned at both ends (E = 200 GPa). The Euler buckling load (in kN, round off to two decimal places) is _____

Q8 MCQ

A pinion of 18 teeth and module 4 mm meshes with a rack. The distance moved by the rack for one complete revolution of the pinion is about

(A) 226.2 mm
(B) 72.0 mm
(C) 113.1 mm
(D) 452.4 mm

Q9 NAT

A disc with a mass moment of inertia of 0.5 kg·m² is fixed to the free end of a vertical steel shaft of 20 mm diameter and 1 m length, whose upper end is fixed (G = 80 GPa). Neglecting the inertia of the shaft, the natural frequency of torsional vibration (in Hz, round off to two decimal places) is _____

Q10 MCQ

A close-coiled helical compression spring has a wire diameter of 5 mm, a mean coil diameter of 40 mm and 12 active coils (G = 80 GPa). Its stiffness is about

(A) 16.3 N/mm
(B) 4.07 N/mm
(C) 8.14 N/mm
(D) 32.6 N/mm

Q11 NAT

A nozzle delivers a water jet of 50 mm diameter at 30 m/s. Taking the density of water as 1000 kg/m³, the kinetic power of the jet (in kW, round off to two decimal places) is _____

Q12 MCQ

A small sphere of diameter D loses heat to an infinite, stagnant fluid by pure conduction only (no fluid motion). The Nusselt number hD/k based on the fluid conductivity is

(A) 1
(B) 2
(C) 3.66
(D) 4.36

Q13 NAT

Steam expands adiabatically in a nozzle with an enthalpy drop of 200 kJ/kg. The velocity at the inlet is negligible. The exit velocity of the steam (in m/s, round off to one decimal place) is _____

Q14 MCQ

Gravity die casting (permanent-mould casting) differs from sand casting mainly in that

(A) the molten metal is forced into the mould under a high pressure
(B) the mould is broken after each casting to remove it
(C) it is best suited for steels with very high melting points
(D) a reusable metal mould is filled under gravity alone

Q15 MCQ

An item has an annual demand of 3600 units, an ordering cost of ₹50 per order and a holding cost of ₹4 per unit per year. The economic order quantity and the number of orders per year are

(A) 300 units and 15 orders
(B) 300 units and 12 orders
(C) 424 units and 8.5 orders
(D) 212 units and 17 orders

Answer Key

Q	Type	Answer
Q1	MCQ	(B) Barren
Q2	NAT	12
Q3	MCQ	(D) 2 cosh x
Q4	NAT	1.476 (accept 1.470 to 1.482)
Q5	NAT	330.2 (accept 328.5 to 331.9)
Q6	MCQ	(C) d/6
Q7	NAT	52.64 (accept 52.2 to 53.1)
Q8	MCQ	(A) 226.2 mm
Q9	NAT	7.98 (accept 7.92 to 8.04)
Q10	MCQ	(C) 8.14 N/mm
Q11	NAT	26.51 (accept 26.31 to 26.71)
Q12	MCQ	(B) 2
Q13	NAT	632.5 (accept 628.0 to 637.0)
Q14	MCQ	(D) a reusable metal mould is filled under gravity alone
Q15	MCQ	(B) 300 units and 12 orders

Detailed Solutions

Q1 **Answer: (B) Barren**

Concept: 'Prolific' means producing a great deal (a prolific inventor, a prolific crop).

Productive, abundant and fertile are all close to it in meaning.

The opposite is **barren** (producing nothing), option (B).

Exam tip: When three options share one idea, the odd one out is usually the antonym.

Q2 **Answer: 12**

Concept: A convex polygon with n sides can be split into (n − 2) triangles from one vertex, so the interior angles add up to (n − 2) × 180°.

(n − 2) × 180 = 1800 gives n − 2 = 10, so n = **12**.

Check: each exterior angle of a regular 12-gon is 360°/12 = 30°, so each interior angle is 150° and 12 × 150° = 1800°.

Exam tip: The exterior angles of any convex polygon always add up to 360°; this is often the quicker route.

Q3 **Answer: (D) 2 cosh x**

Concept: The auxiliary equation $m^2 - 1 = 0$ gives $m = \pm 1$, so $y = Ae^x + Be^{-x}$.

$y(0) = A + B = 2$ and $y'(0) = A - B = 0$, so $A = B = 1$ and $y = e^x + e^{-x} =$ **2 cosh x**, option (D).

2 cos x solves $y'' + y = 0$ instead, and 2 sinh x has $y(0) = 0$.

Exam tip: Check the initial conditions on each option directly; it is often faster than solving.

Q4 **Answer: 1.476 (accept 1.470 to 1.482)**

Concept: Simpson's 1/3 rule: $\int f\,dx \approx (h/3)[f_0 + 4f_1 + f_2]$ for two intervals.

$f(0) = 1$, $f(0.5) = e^{0.25} = 1.2840$, $f(1) = e = 2.7183$.

$I \approx (0.5/3)[1 + 4 \times 1.2840 + 2.7183] = 0.16667 \times 8.8544 =$ **1.476** (the exact value is 1.4627).

Exam tip: Simpson's 1/3 rule needs an even number of intervals; with an odd number use the 3/8 rule for the last three.

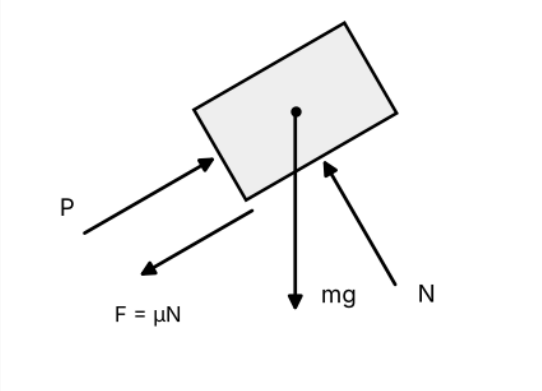
Q5 **Answer: 330.2 (accept 328.5 to 331.9)**

Concept: At constant speed the forces along the plane balance: $P = mg \sin\theta + \mu mg \cos\theta$ (friction acts down the plane, opposing the motion).

$mg = 50 \times 9.81 = 490.5$ N. Normal reaction $N = 490.5 \cos 30° = 424.8$ N, friction $= 0.2 \times 424.8 = 85.0$ N.

$P = 490.5 \times 0.5 + 85.0 = 245.25 + 84.96 =$ **330.2 N**.

Exam tip: For motion down the plane at constant speed the friction term changes sign: $P = mg(\sin\theta - \mu\cos\theta)$.



Q6 **Answer: (C) d/6**

Concept: Direct stress P/A and bending stress Pe/Z combine; the far edge stays free of tension while $P/A \geq Pe/Z$, i.e. $e \leq Z/A$.

For bending about the axis parallel to b: $Z = bd^2/6$ and $A = bd$, so $Z/A =$ **d/6**, option (C).

The load must therefore lie within the 'middle third' of the depth (d/6 on either side of the centroid).

Exam tip: For a solid circular section the core is a circle of diameter d/4 ($e \leq d/8$).

Q7 **Answer: 52.64 (accept 52.2 to 53.1)**

Concept: Euler's load for a pin-ended strut is $P_{cr} = \pi^2 EI/L^2$, and the strut buckles about the axis with the **least** moment of inertia.

$I_{min} = 40 \times 20^3/12 = 26\,667$ mm⁴ (the other value, $20 \times 40^3/12 = 106\,667$ mm⁴, does not govern).

$P_{cr} = \pi^2 \times 200\,000 \times 26\,667/1000^2 = 52\,638$ N = **52.64 kN**.

Exam tip: Using the larger I would give four times the load (210.6 kN), a common trap.

Q8 **Answer: (A) 226.2 mm**

Concept: The pitch line of the rack rolls without slipping on the pitch circle of the pinion.

Pitch circle diameter $d = mz = 4 \times 18 = 72$ mm. One revolution moves the rack by the pitch circumference $\pi d = \pi \times 72 =$ **226.2 mm**, option (A).

Equivalently, 18 teeth × circular pitch πm (12.57 mm) = 226.2 mm.

Exam tip: 72 mm is the pitch diameter and 113.1 mm is only half a revolution.

Q9 **Answer: 7.98 (accept 7.92 to 8.04)**

Concept: The torsional stiffness of a shaft is $k_t = GJ/L$, and $f_n = (1/2\pi)\sqrt{(k_t/I)}$.

$J = \pi \times 0.02^4/32 = 1.5708 \times 10^{-8}$ m⁴, so $k_t = 80 \times 10^9 \times 1.5708 \times 10^{-8}/1 = 1256.6$ N·m/rad.

$\omega_n = \sqrt{(1256.6/0.5)} = 50.13$ rad/s, $f_n = 50.13/2\pi =$ **7.98 Hz**.

Exam tip: This is the rotational twin of $\sqrt{(k/m)}$: replace k by GJ/L and m by I.

Q10 **Answer: (C) 8.14 N/mm**

Concept: For a close-coiled helical spring, $k = Gd^4/(8D^3n)$.

$k = 80\,000 \times 5^4/(8 \times 40^3 \times 12) = 80\,000 \times 625/6\,144\,000 =$ **8.14 N/mm**, option (C).

Note the strong dependence: doubling d multiplies k by 16, while doubling D divides it by 8.

Exam tip: Cutting a spring into two halves doubles the stiffness of each half (n is halved).

Q11 **Answer: 26.51 (accept 26.31 to 26.71)**

Concept: Power of a jet = (mass flow rate) × (kinetic energy per kg) = $(\rho AV) \times V^2/2 = \frac{1}{2}\rho AV^3$.

$A = (\pi/4) \times 0.05^2 = 1.9635 \times 10^{-3}$ m². $\dot{m} = 1000 \times 1.9635 \times 10^{-3} \times 30 = 58.9$ kg/s.

$P = 0.5 \times 58.9 \times 30^2 = 26\,507$ W = **26.51 kW**.

Exam tip: Jet power grows with V^3, so doubling the jet velocity (same area) multiplies it by eight.

Q12 **Answer: (B) 2**

Concept: Treat the fluid as a spherical shell from r = D/2 to r → ∞. Its conduction resistance is $(1/r_1 - 1/r_2)/(4\pi k) \to 1/(4\pi k\, r_1)$.

So $Q = 4\pi k\, r_1 \Delta T$. Writing $Q = h(4\pi r_1^2)\Delta T$ gives $h = k/r_1 = 2k/D$.

Hence $Nu = hD/k =$ **2**, option (B). This is the lower limit of every forced- or free-convection correlation for spheres (e.g. $Nu = 2 + 0.6Re^{1/2}Pr^{1/3}$).

Exam tip: 3.66 and 4.36 are the fully developed laminar values for flow inside a tube (constant wall temperature and constant heat flux).

Q13 **Answer: 632.5 (accept 628.0 to 637.0)**

Concept: Steady-flow energy equation for an adiabatic nozzle with no work: $h_1 + V_1^2/2 = h_2 + V_2^2/2$.

With $V_1 \approx 0$: $V_2 = \sqrt{(2\Delta h)} = \sqrt{(2 \times 200\,000)}$ m/s (Δh in J/kg).

$V_2 = \sqrt{400\,000} =$ **632.5 m/s**.

Exam tip: Remember $V = 44.72\sqrt{\Delta h}$ with Δh in kJ/kg; forgetting to convert kJ to J gives a value 31.6 times too small.

Q14 **Answer: (D) a reusable metal mould is filled under gravity alone**

Concept: In gravity die casting the mould is made of metal (usually cast iron or steel) and is reused for many castings; the metal enters under its own head only.

So the answer is **a reusable metal mould filled under gravity**, option (D).

Forcing metal in under pressure is pressure die casting; breaking the mould describes sand and investment casting; and the process suits lower-melting alloys such as aluminium, magnesium and copper alloys.

Exam tip: Metal moulds chill the casting, giving finer grains and better finish but needing low-melting alloys.

Q15 **Answer: (B) 300 units and 12 orders**

Concept: EOQ = $\sqrt{(2DC_o/C_h)}$, which balances the annual ordering and holding costs.

EOQ = $\sqrt{(2 \times 3600 \times 50/4)} = \sqrt{90\,000} = 300$ units.

Number of orders = D/EOQ = 3600/300 = 12 per year, so the answer is **300 units and 12 orders**, option (B).

Exam tip: At the EOQ both costs are equal: (3600/300) × 50 = ₹600 = (300/2) × 4.

What next?

Day 90 test tonight, 7 to 11 PM

65 questions · 180 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

Website test series

Dig Career Thrust app

Telegram: daily PDFs

That's the final day of the challenge. All the best for GATE! Comment **GATE90** on our reel to get all links.
Want concise revision notes? The GATE ME handwritten short notes of an AIR 16 topper are available in the Dig Career Thrust app.